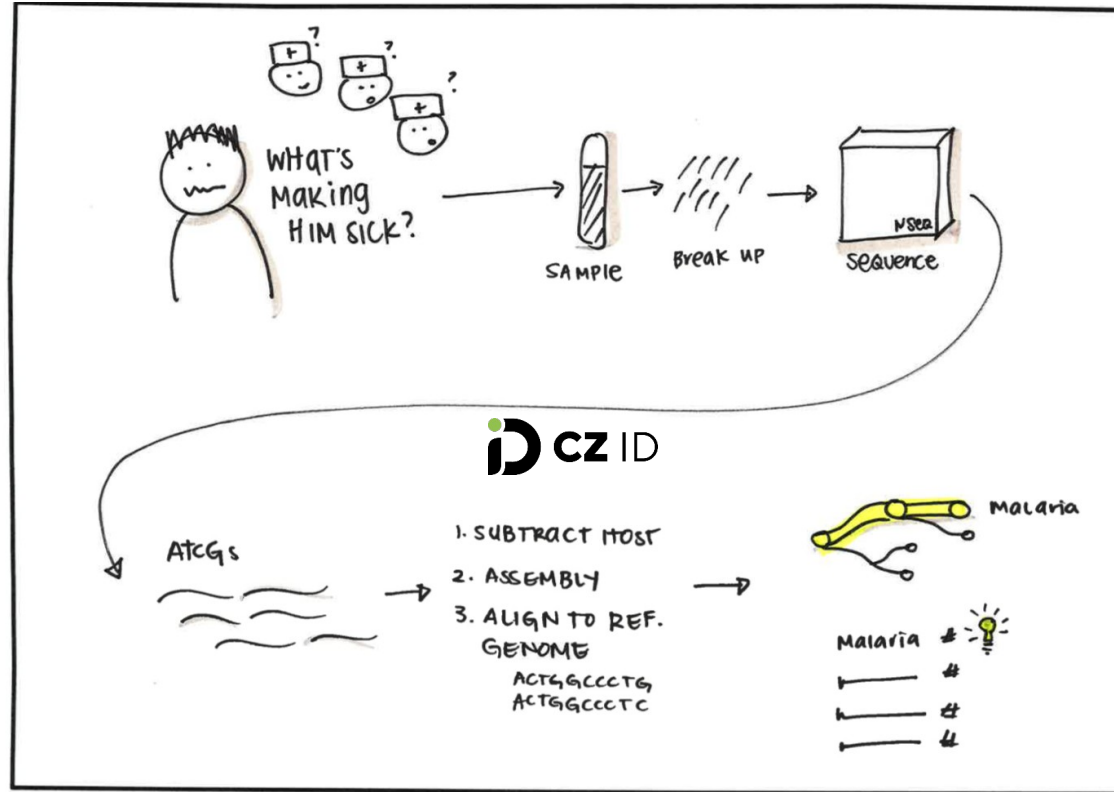




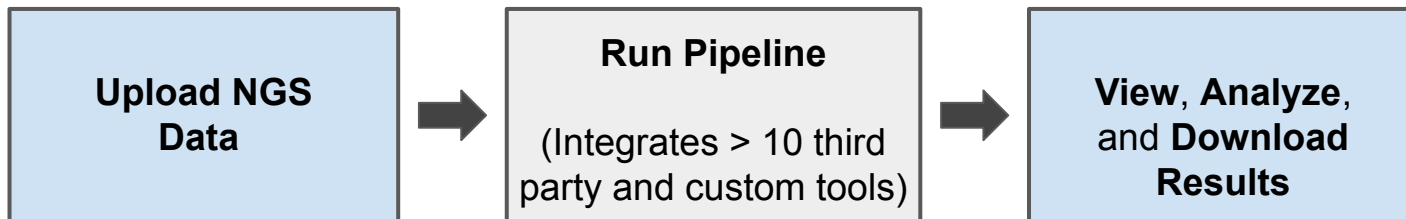
Introduction to CZ ID

CZ ID is a free platform used to identify microbes of interest from NGS data



CZ ID integrates bioinformatic tools to enable NGS analysis regardless of available computational resources

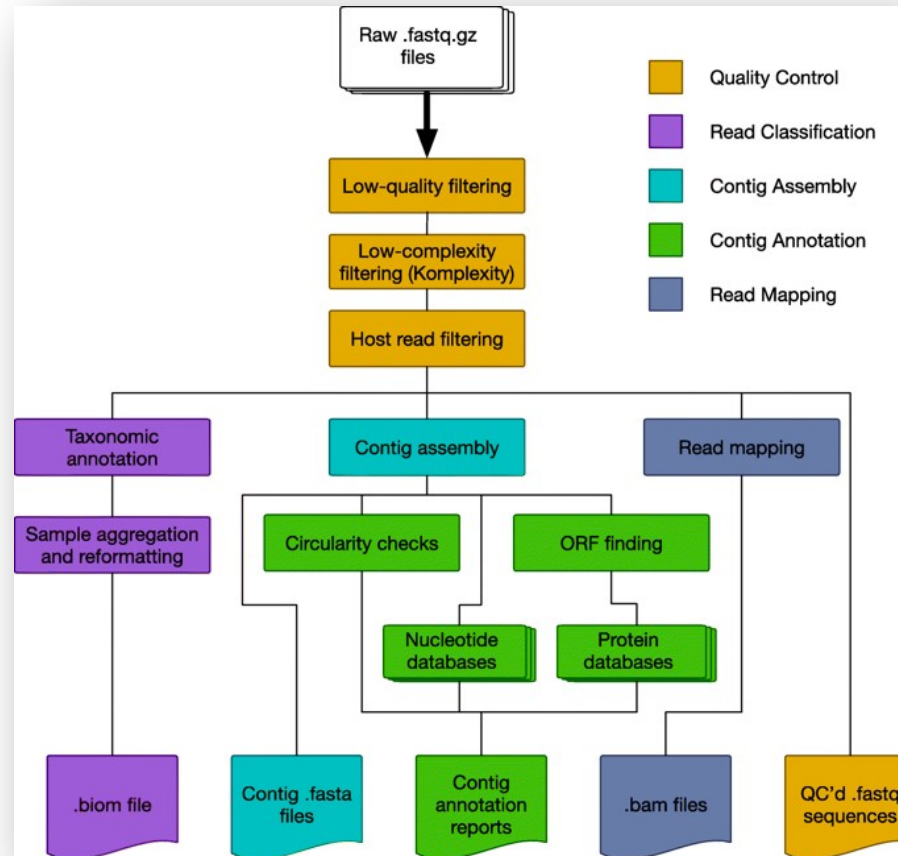
General Workflow:



Plug-and-play, cloud-based platform

(optimized for scalable Amazon Web Services (AWS) cloud deployment)

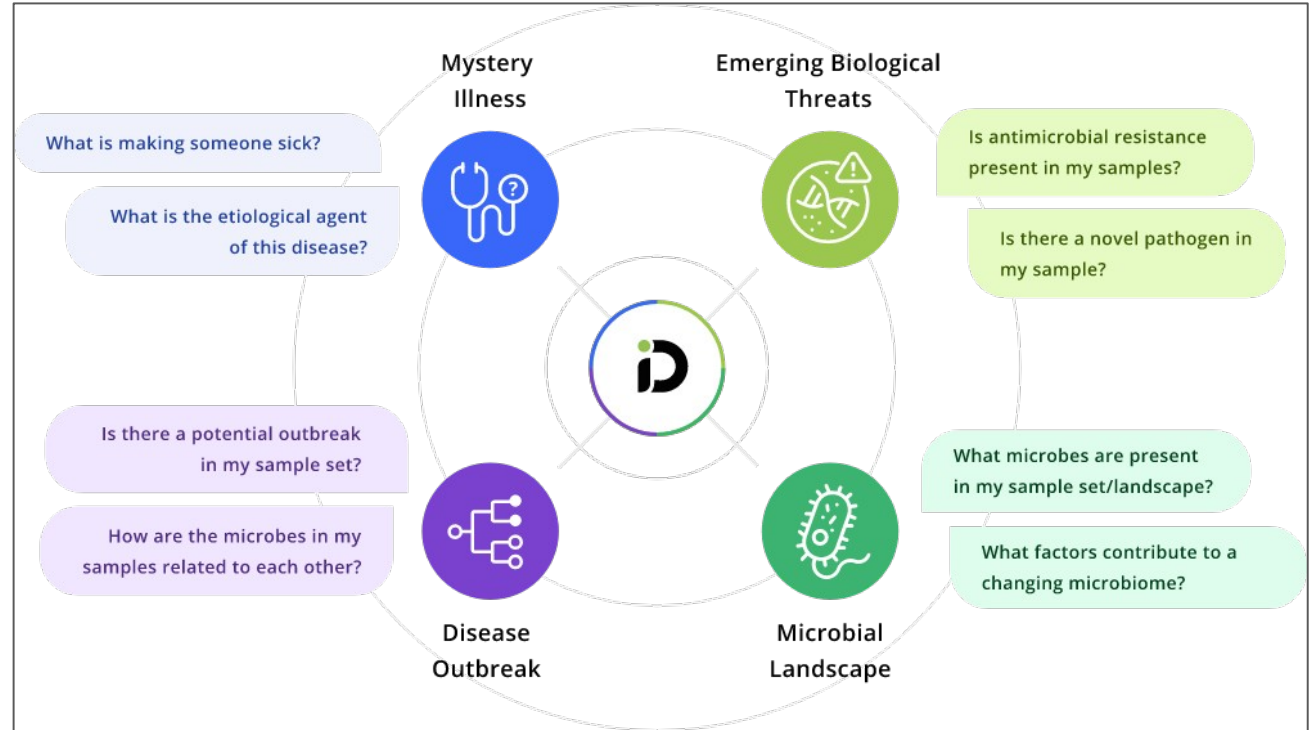
NGS data analysis requires computational power and includes MANY steps



CZ ID supports multiple uses

CZ ID Pipelines:

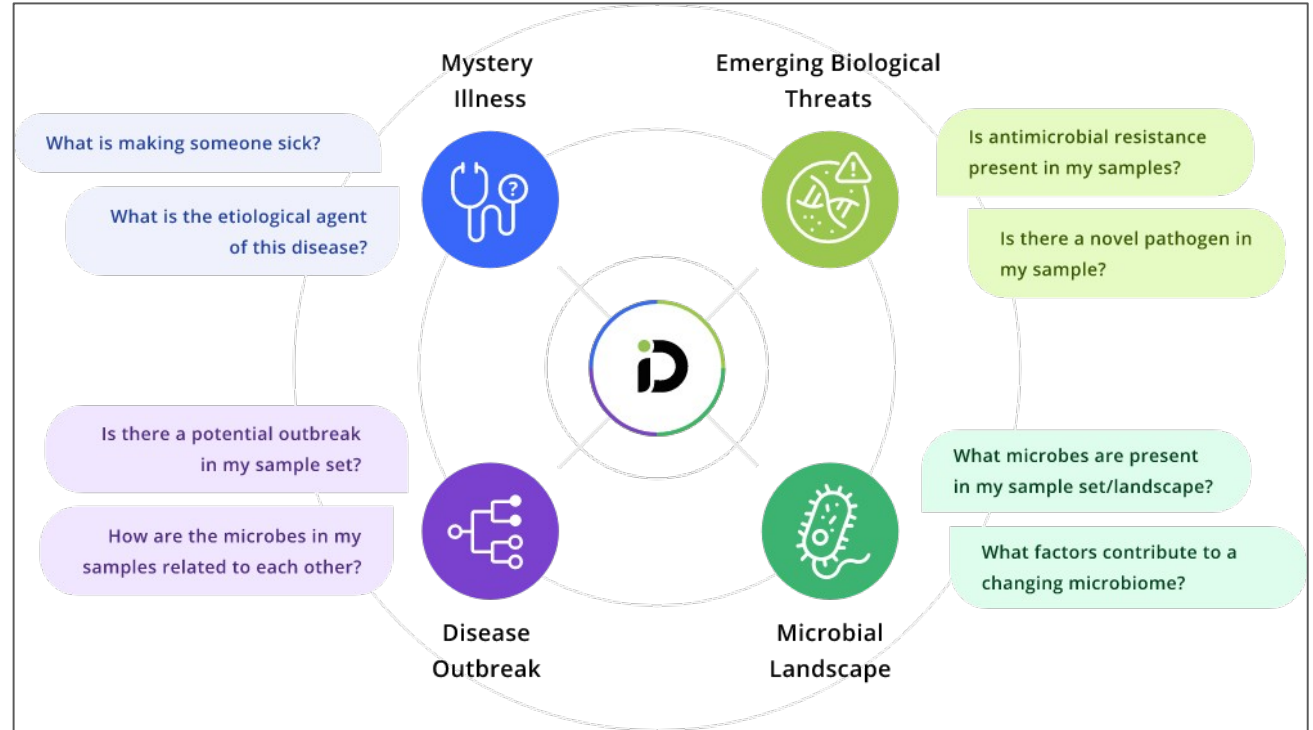
- Metagenomic (mNGS) analysis
- Antimicrobial Resistance (AMR) gene detection
- Viral consensus genome assembly



CZ ID supports multiple uses

CZ ID Pipelines:

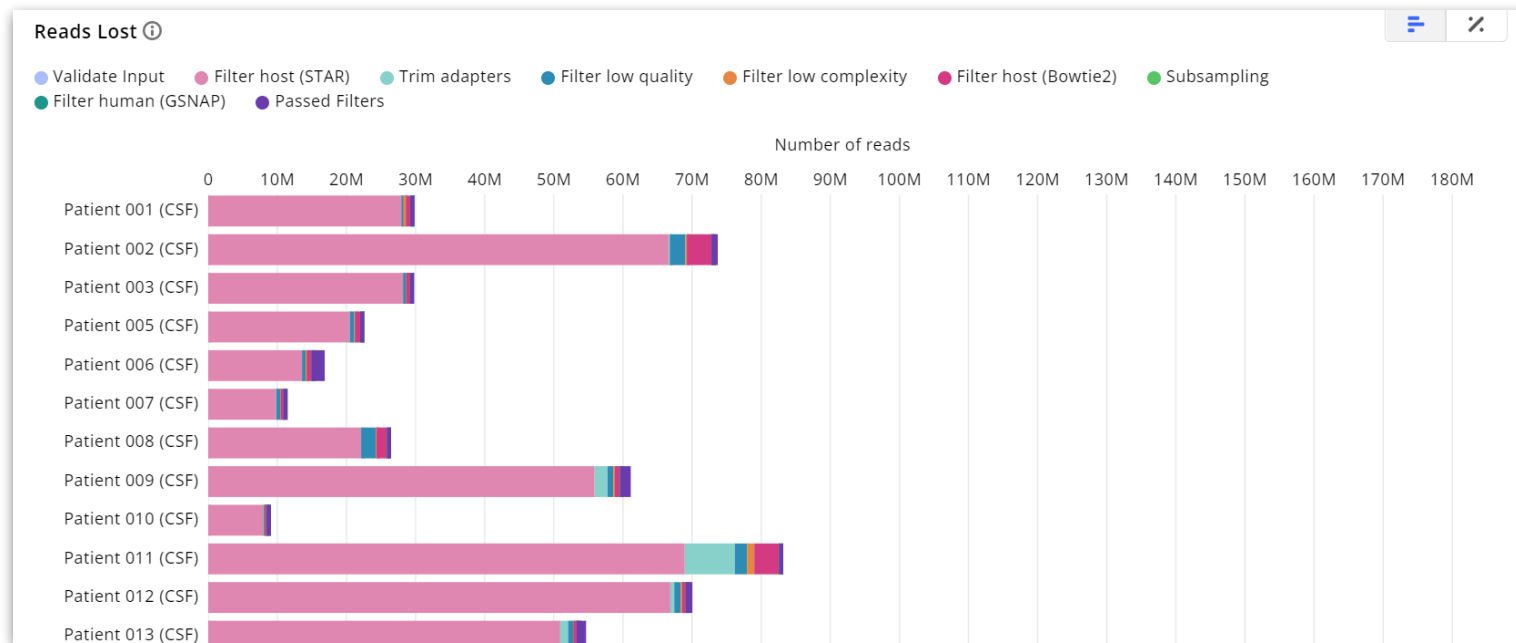
- **Metagenomic (mNGS) analysis**
- Antimicrobial Resistance (AMR) gene detection
- Viral consensus genome assembly



CZ ID mNGS Analysis Highlights

- With just a few clicks, easily:
 - Evaluate sample quality
 - Explore and evaluate identified taxa through interactive reports
 - Compare samples using interactive heatmaps
 - Generate viral consensus genomes
 - Keep track of pipeline steps

Perform QC



Explore and evaluate metrics for detected microbes

< Medical Detectives

Patient 010 (CSF) ▾

Share

Download ▾

?

...

Metagenomic

Antimicrobial Resistance (Deprecated)

Consensus Genome

Taxon name

Q

Name Type: Scientific ▾

Background: MedicalDetectives_WaterCtrls ▾

Categories ▾

Threshold filters: 1 ▾

Read Specificity: Specific Only ▾

Annotation ▾

NT rPM >= 10 ✕

112 rows passing the above filters, out of 3100 total rows

CLEAR FILTERS

Table

NT NR

>	Taxon	Score ▾	Z ...	rPM	r	con...	con...	%...	L	E v...	NT NR
▾	Alphavirus (1 viral species: ● 1)	72,976,252	100.0 100.0	3,650.9 3,646.7	19,344 19,322	1 1	19,139 19,139	100.0 99.9	11,671.6 2,450.2	10 ⁻³⁰⁵ 10 ⁰	
	Chikungunya virus Known Pathogen	72,976,252	100.0 100.0	3,650.9 3,646.7	19,344 19,322	1 1	19,139 19,139	100.0 99.9	11,671.6 2,450.2	10 ⁻³⁰⁵ 10 ⁰	
>	Tetraparvovirus (1 viral species:)	46,706,236	100.0 100.0	2,366.2 2,304.6	12,537 12,211	2 1	12,102 11,987	99.1 99.6	4,677.3 896.3	10 ⁻²⁹⁶ 10 ⁰	
>	Providencia (1 bacterial species: ● 1)	14,332,531	100.0 0.0	1,433.3 0.0	7,594 0	1 0	7,594 0	99.8 0.0	1,354.0 0.0	10 ⁻³⁰⁸ 0	
>	Lactobacillus (1 bacterial species:)	2,502,625	100.0 100.0	28.9 293.5	153 1,555	7 9	79 1,475	98.6 78.8	209.0 157.7	10 ⁻¹⁰⁸ 10 ⁻¹	
>	Rothia (1 bacterial species:)	2,364,849	100.0 100.0	238.4 5.5	1,263 29	6 0	1,127 0	96.5 97.3	2,086.8 45.7	10 ⁻²⁷² 10 ⁻¹⁶	

Example

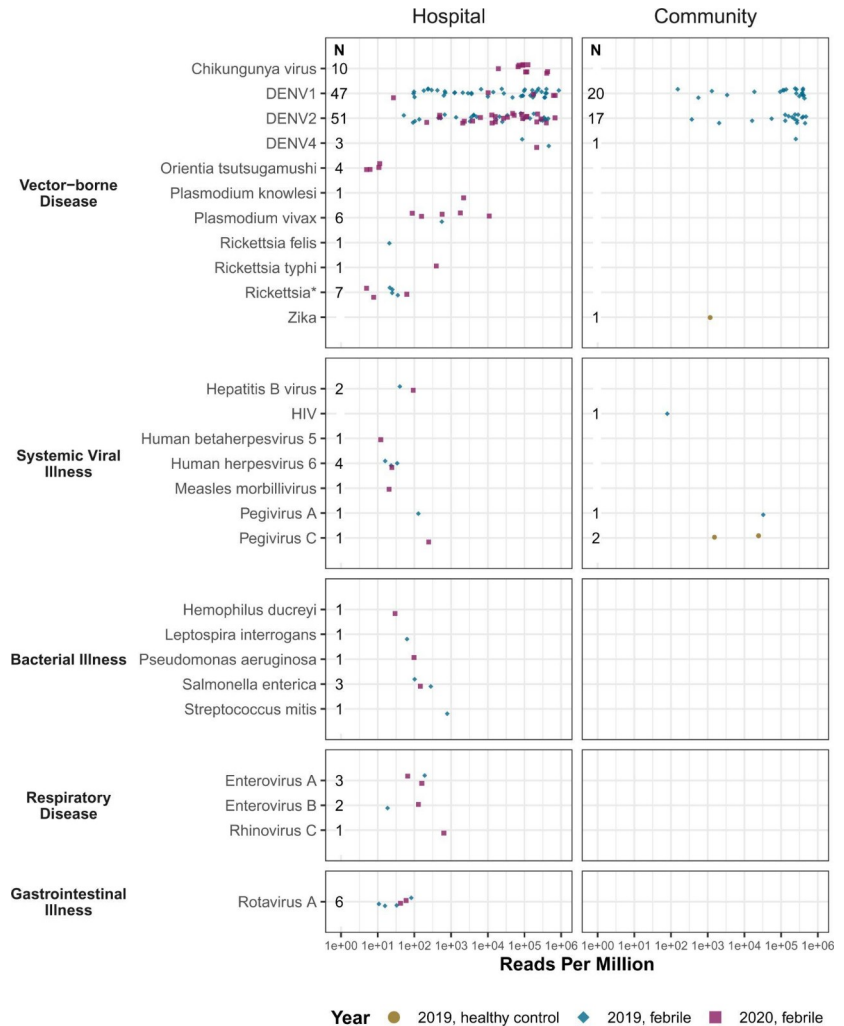
Study: Pathogen landscape associated with febrile illness in Cambodia

Sample type: Serum

Results: 203 pathogens detected from 489 samples

Figure: Pathogens organized by disease category. Detection distinguished by participant cohort (community- (n = 87) vs hospital-based cohorts (n = 377))

[Bohl et al 2021](#)



Example

Study: Pathogen landscape associated with pediatric febrile illness in Tororo, Uganda

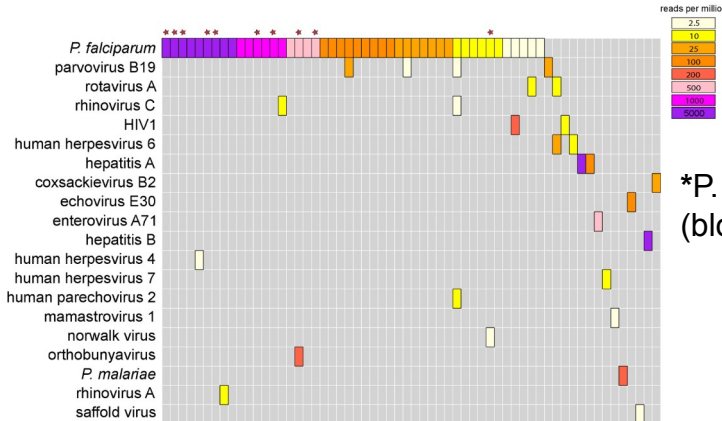
Sample type: Serum and Nasopharyngeal (NP) swabs

Results: Pathogens detected in more than half of serum and NP samples.

Figure: Pathogens organized by sample type

[Ramesh et al 2019](#)

Serum

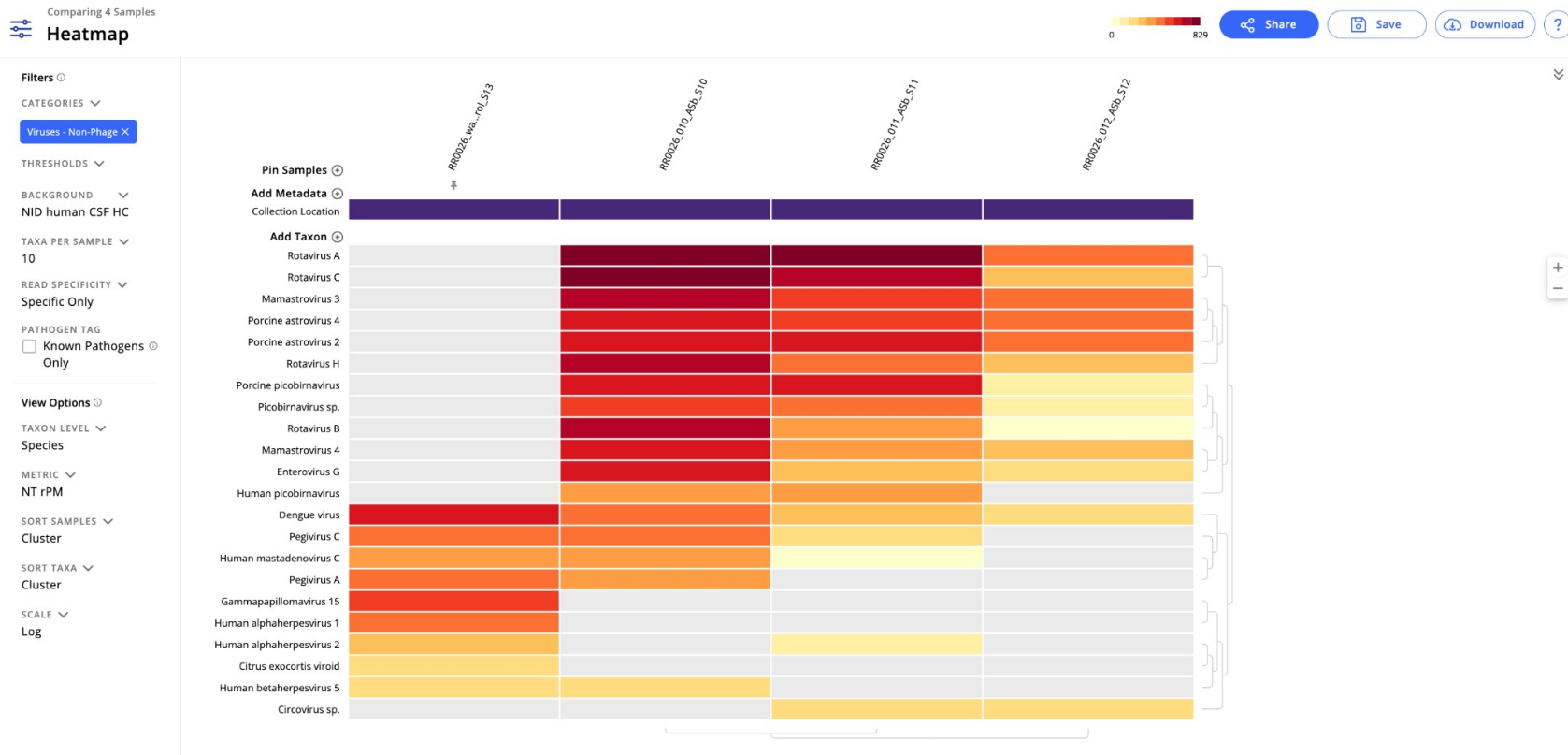


**P. falciparum* +
(blood smear test)

NP Swabs



Compare samples visually to identify patterns of interest



Create viral consensus genomes

Is my consensus genome complete? ⓘ

Taxon	Mapped Reads	GC Content	SNPs	%id	Informative Nucleotides	% Genome Called	Missing Bases	Ambiguous Bases
Chikungunya virus	1048870	50.1%	69	99.4%	11713	100%	1	0

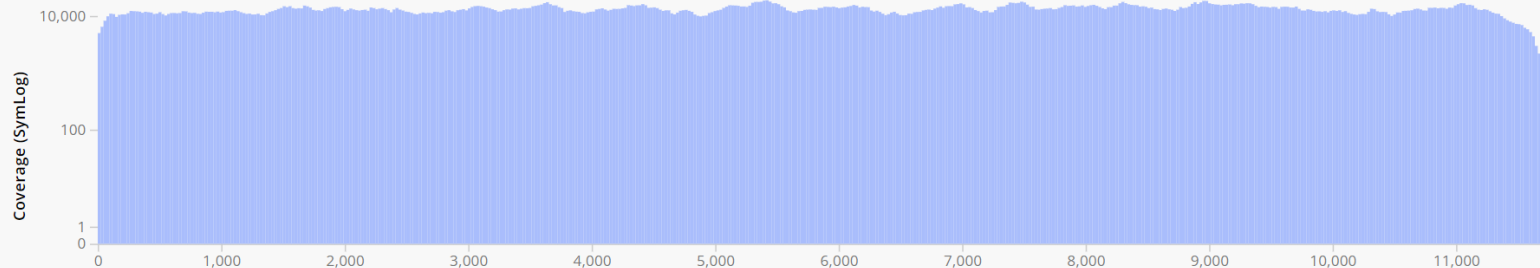
How good is the coverage? ⓘ

Reference NCBI Entry [KX619426.1](#)

Reference Length 11708

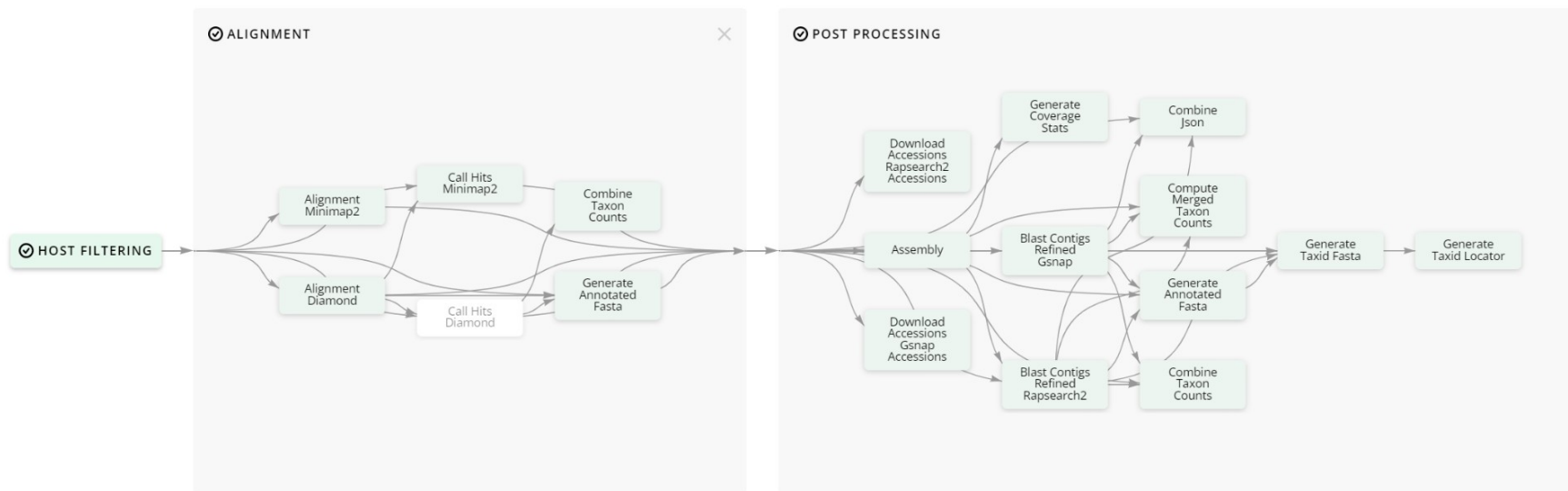
Coverage Depth 13075.0x

Coverage Breadth 100.0%



KX619426.1 - Chikungunya virus isolate JHCK128 nonstructural polyprotein and structural polyprotein genes, complete cds

View pipeline steps and access intermediate files



Pipeline visualization

And more!

Let's look at the pipeline...